

Recurrence Sites and Contraindications for C&D

Recurrences of basal cell carcinoma (BCC) are most frequently observed on the forehead, temple, ears, nose, and shoulders. **Curettage and desiccation (C&D)** is not recommended for large primary BCC, morpheaform BCC, or recurrent BCC due to higher recurrence rates and difficulty in assessing margins.

Cryosurgery (See Chap. 247)

Cryosurgery is a destructive modality used in the treatment of BCC. Two freeze-thaw cycles achieving a tissue temperature of $-50^{\circ}\mathrm{C}$ ($-58^{\circ}\mathrm{F}$) are necessary to effectively destroy BCC cells. A margin of clinically normal-appearing tissue should also be treated to address subclinical extension.

A key limitation of cryosurgery is the absence of histologic confirmation of complete tumor removal. Despite this, Kuflik and Gage reported a 99% cure rate in 628 patients followed for 5 years.

Possible complications include: - Hypertrophic scarring - Post-inflammatory pigmentary changes - Fibrous scar tissue that may obscure tumor recurrence

Any new change in a cryosurgery-treated area after complete healing should raise suspicion for recurrent BCC.

Topical Treatment

Imiquimod

Imiquimod 5% cream (see Chap. 221) is a Toll-like receptor 7 agonist that induces interferon- α and other cytokines, enhancing T helper 1-type immunity.

Clinical and histologic clearance rates: - **Superficial BCC**: 75% (5 times/week) and 73% (7 times/week) in double-blind, vehicle-controlled trials - **Nodular BCC**: Approximately 53% (10 of 19 lesions cleared in one study)

Adverse effects are typically limited to local skin reactions (e.g., erythema, erosion, crusting). Follow-up durations in available case series are generally shorter than those for surgical treatments, limiting long-term recurrence data.

5-Fluorouracil (See Chap. 220)

5-Fluorouracil (5-FU), commonly used for actinic keratoses, has also been employed in BCC treatment.

- Epstein reported a **5-year recurrence rate of 21%** after 5-FU monotherapy, which decreased to **6%** when combined with initial curettage.
- 5-FU is metabolized by dihydropyrimidine dehydrogenase; therefore, it is contraindicated in patients with deficiency in this enzyme.

Due to variable efficacy and risk of treatment failure, use of 5-FU for BCC should be carefully considered, with attention to lesion characteristics and recurrence risk.

Photodynamic Therapy (See Chap. 239)

Photodynamic therapy (PDT) uses a photosensitizing agent activated by visible light to generate cytotoxic oxygen species that destroy cancer cells. Topical δ -aminolevulinic acid (ALA) is commonly used to increase intracellular accumulation of protoporphyrin IX, which selectively builds up in tumor cells.

- Morrison et al. reported an **88% initial clearance rate** in 40 large BCCs (>2 cm) after 1–3 treatments, with follow-up of 12–60 months.
- Recurrence rates vary widely across studies, with Marmur et al. reporting BCC recurrence rates between **0% and 31%**.

Given the potential for higher recurrence rates compared to surgical methods, PDT is best reserved for selected cases where standard therapies are not feasible.

Radiation Therapy (See Chap. 241)

Radiation therapy (XRT) may be considered for: - Primary BCC in patients unsuitable for surgery - Cases with positive surgical margins

Advantages: - Minimal patient discomfort - Non-invasive option for medically frail or surgery-averse patients

Disadvantages: - No histologic confirmation of tumor clearance - Prolonged treatment course - Cosmetic outcomes may deteriorate over time - Risk of aggressive, extensive recurrences if tumor persists or returns

Reported local control rates range from **93% to 97%**, though cosmetic results are generally rated inferior to those achieved with surgery.

Special Management Issues

Incompletely Excised Basal Cell Carcinoma

Historical data from the 1960s suggest that **50% to 70% of incompletely excised BCCs do not recur**. Based on this, some clinicians have adopted a "wait and see" approach, monitoring the site closely for signs of recurrence rather than proceeding immediately to further intervention. This strategy requires careful patient selection and long-term follow-up.